



THEREMIN

Circuit

Using a 9V dc source and **Pure Analog** implementation

IN SHORT:-

“The circuit uses a CD4093-based RC oscillator with a potentiometer for frequency control, an op-amp for mixing and amplification, and an LM386 to drive an 8Ω speaker.”

COMPONENTS:-

CD4093

MCP602

LM386

Resistors (5.1k, 10k, 100k, 1M)

Capacitors (100pF, 0.1 μ F, 10 μ F)

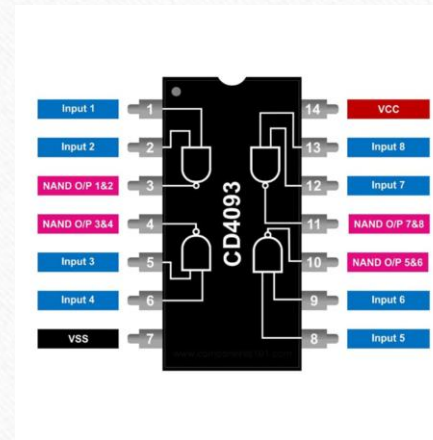
10k Potentiometer

Antenna

8 Ω Speaker

7805 Ic

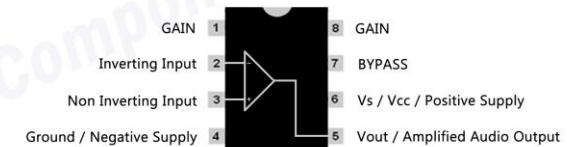
9V dc battery



LM386 IC Pinout



LM386 DETAILED PIN DESCRIPTION



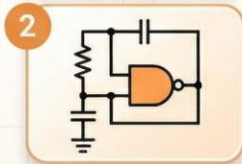
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Electronics Components Uses, Features, Pinouts, Equivalents, Applications & More...

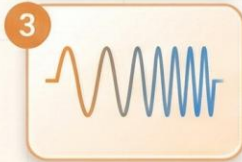
Core Analog Concepts Used



1 Capacitance (Human body coupling)
The Theremin antenna uses the hand to change circuit capacitance.

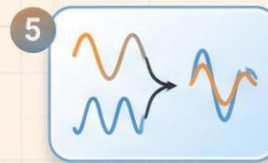


2 RC Time Constant & Oscillation
The Theremin converts the simple circuit & oscillance.



3 Frequency Variation / Modulation
The varforey variation/frequency/frequency in amoured.

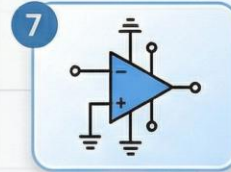
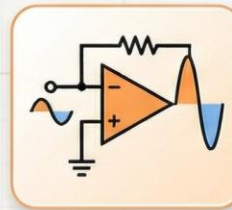
4 Heterodyning (Beat Frequency Generation)



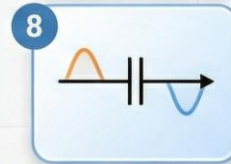
4 Heterodyning
Generation of signals to time overage

5 Signal Mixing (Nonlinear behavior)

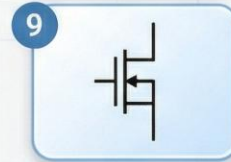
6 Operational Amplifier (Amplification)



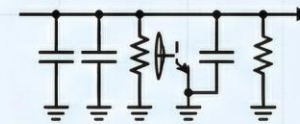
7 Virtual Ground (Midpoint biasing) uss op-amp buffer.



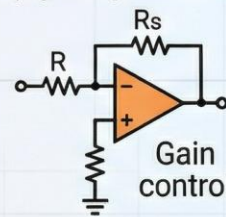
8. AC Coupling (Capacitor signal transfer)



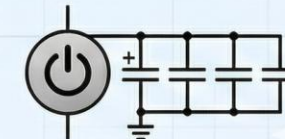
9. Impedance & High-impedance sensing
Decoupling / Noise Filtering



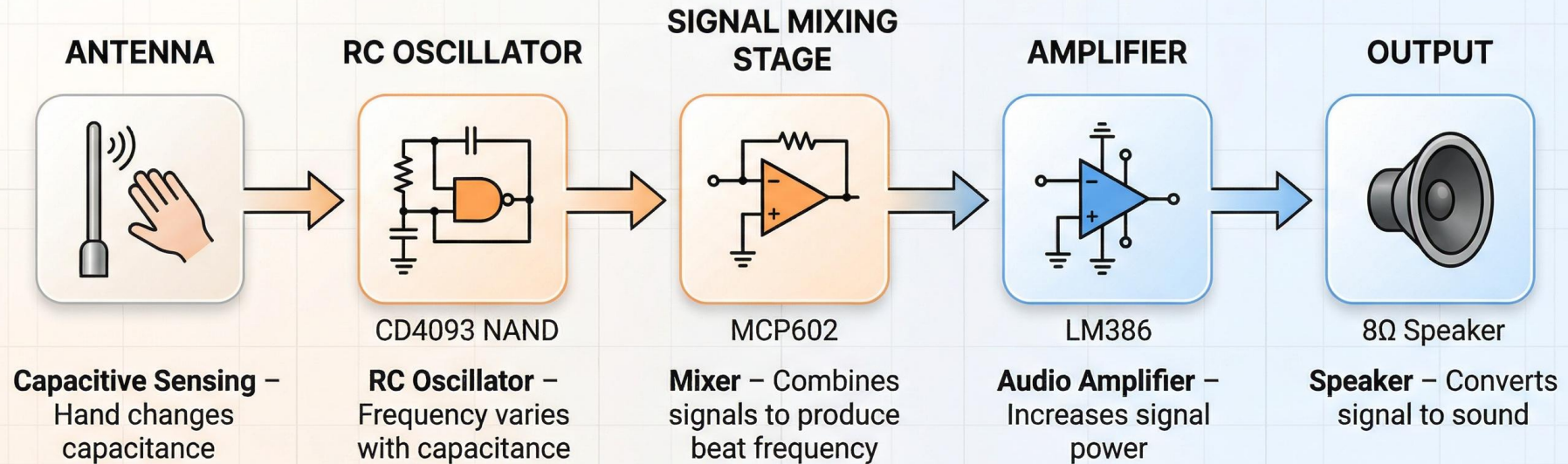
11. Gain & Feedback (Op-amp control)



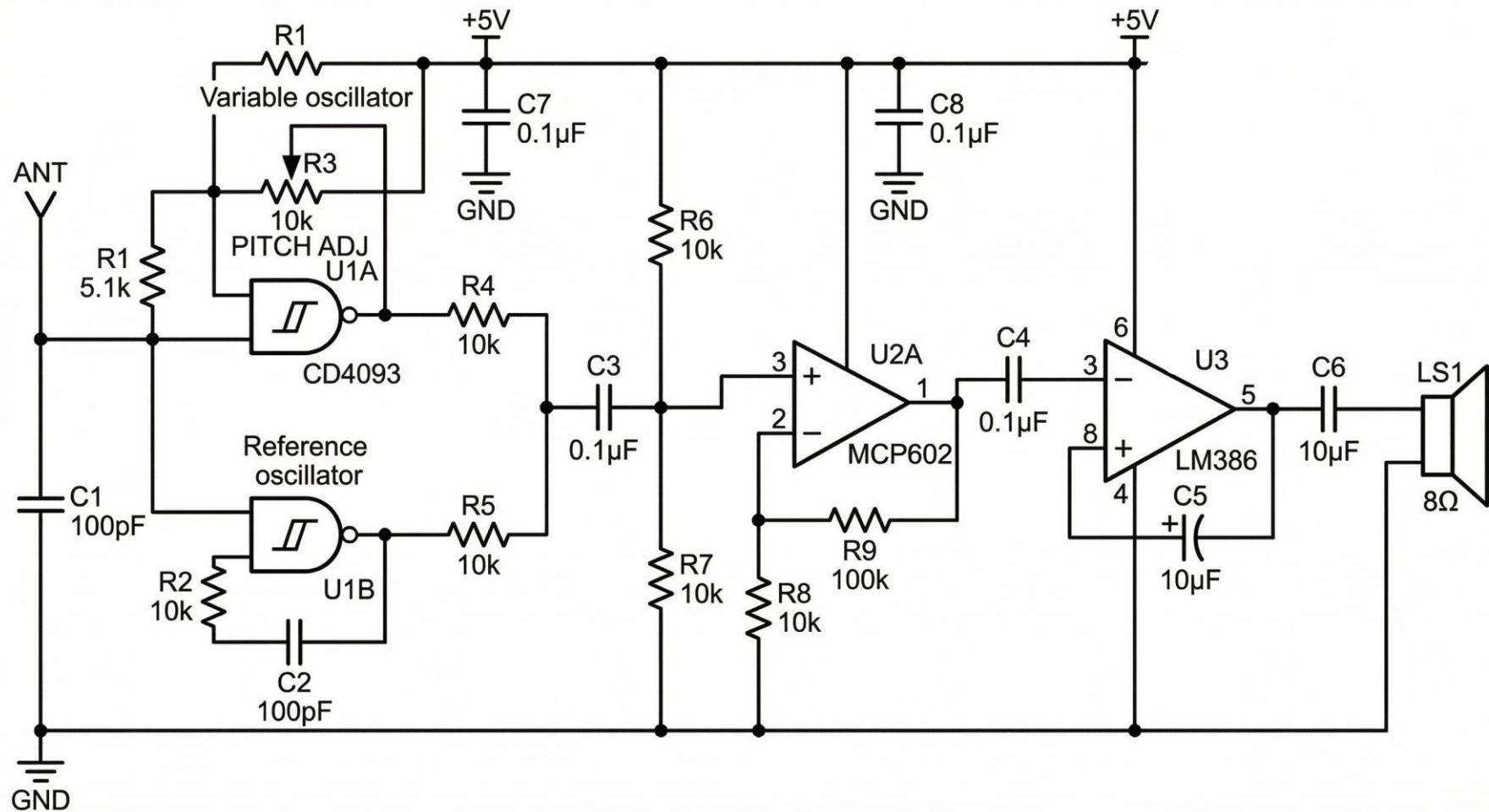
12. Power Supply Stabilization



How a Theremin Circuit Works



Circuit Diagram:-



THE THEREMIN:-

